NAMMA SPACE / TECHFEST 2026-27

StarDust

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Links & Prototype

Round 1: 3D Reconstruction & Roaming

01 Live web viewer

> https://stardust-ns.vercel.app/

Open on a desktop or laptop browser. Wait for loading, then click **Start walkthrough**. Use **WASD** or arrow keys to walk, the **mouse** to look, **Shift** to move faster, **Q/E** to adjust eye height, **R** to reset and **Esc** to release the mouse.

Current prototype: full room2 reconstruction, all 1,474 registered frames at 1600×900, 60,000 training steps; 1,296,805 exported Gaussians and a 29.41 MiB SPZ model.

02 Code repository & reconstruction setup

> https://github.com/positron-nb/starDust_NS

Branch: **main**. `README.md` contains viewer setup and reconstruction instructions; `capture_sop.md` describes data collection. The submitted model was trained locally in WSL2 on an RTX 5060. `scripts/train_file_backed.py` provides the memory-mapped image cache; `docs/local_full_training.md` documents the tested settings. The Colab notebook remains an alternative.

Sample data: `test/sample_data/`. Usage: `test/README.md`.

03 Submission folder

> Open the Round 1 Google Drive folder

Technical abstract: **01_Technical_Abstract.pdf** - system architecture, capture SOP, full-frame reconstruction results and updated screenshots. The folder also includes the current source-code ZIP and the model/run-output ZIP.